

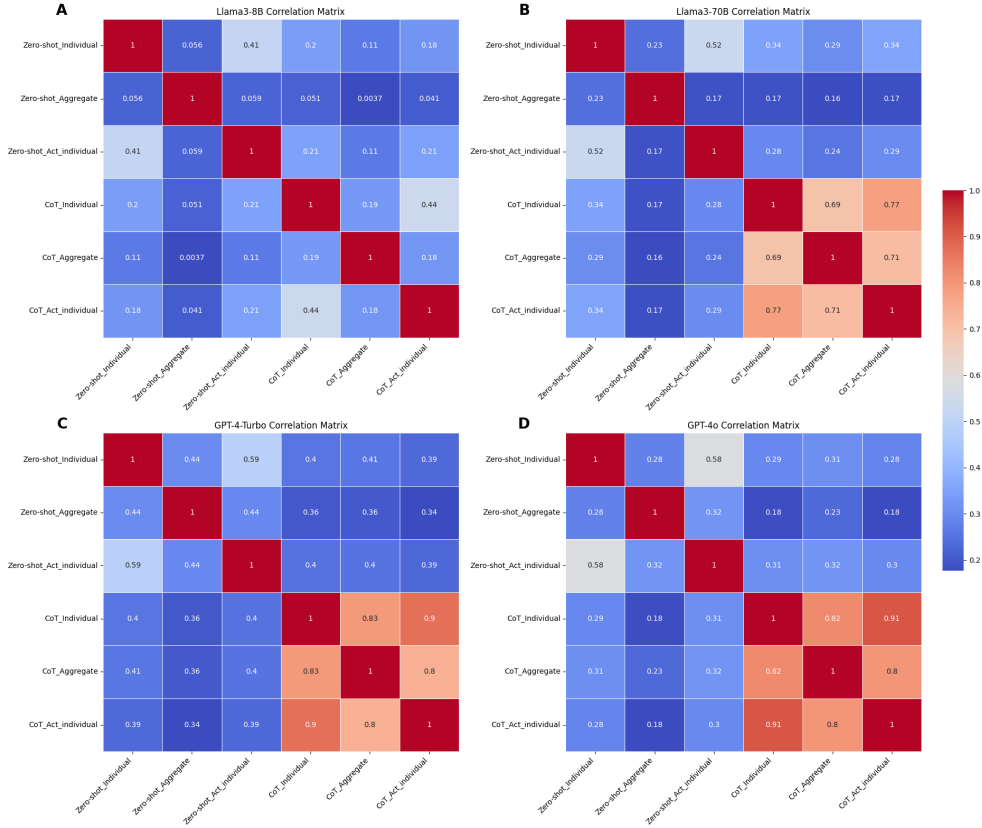


Figure 2: The correlations between LLMs [Llama3-8B, Llama3-70B, GPT-4-Turbo (0125-preview), GPT-4o]

weighting functions (e.g., like the one in Prospect Theory). Notably, one of the probability weighting functions is usually linear, and the other S-shaped. In the present case, one of the weighting functions was linear, but the other approximated a flat line. This suggests that GPT-4 expected people to be approximately rational most of the time, but completely ignores probabilities (i.e., weights them equally) in a minority of cases.

## E ABLATION STUDY FOR FORWARD MODELING

### E.1 DIFFERENT TEMPERATURES

Originally, we chose temperature = 1 for some tasks and temperature = 0.7 for others. While 1 is the default temperature used by most models, 0.7 is another common option, used by works such as AlpacaEval (Li et al., 2023) on instruction following tasks.

To further validate the robustness of our findings, we test the forward modeling experiment using GPT-4o with temperatures 0.0, 0.5, 1.0, 1.5, and 2.0 for both zero-shot and CoT conditions, and compute spearman correlation with both human and max expected value, shown in Tables 12 and 13. We find that varying temperature does not substantially alter our findings other than when conducting CoT with high temperatures, where models become unable to circle back to an answer due to noisy outputs.

Table 11: MSE between GPT-4-individual-zero-shot outputs and the fitted predictions of behavioral models.

| Behavioral Model                        | MSE            |
|-----------------------------------------|----------------|
| Better Than Average                     | 0.20473        |
| Equiprobable                            | 0.20212        |
| Low Payoff Elimination                  | 0.18248        |
| Low Expected Payoff Elimination         | 0.18383        |
| Probable                                | 0.20559        |
| Minimax                                 | 0.20751        |
| Maximax                                 | 0.20401        |
| Priority Heuristic                      | 0.18994        |
| Disappointment Theory with EV           | 0.16125        |
| Disappointment Theory with EU           | 0.12273        |
| Disappointment Theory Without Rescaling | 0.16134        |
| RegretTheory with EV                    | 0.15918        |
| RegretTheory with EU                    | 0.12278        |
| Expected Value                          | 0.16134        |
| Expected Utility                        | 0.11435        |
| Prospect Theory                         | 0.11427        |
| Transfer of Attention Exchange          | 0.12028        |
| Mixture of Theories                     | <b>0.09835</b> |

Table 12: Comparison of Zero-shot vs. Max Expected Value and CoT vs. Max Expected Value across temperature settings for GPT-4o.

| Zero-shot vs. Max Expected Value |          | CoT vs. Max Expected Value |                     |
|----------------------------------|----------|----------------------------|---------------------|
| Temp                             | Spearman | Temp                       | Spearman            |
| 0.0                              | 0.313    | 0.0                        | 0.540               |
| 0.5                              | 0.272    | 0.5                        | 0.515               |
| 1.0                              | 0.284    | 1.0                        | 0.539               |
| 1.5                              | 0.302    | 1.5                        | 0.351               |
| 2.0                              | 0.308    | 2.0                        | <i>model unable</i> |

## E.2 DEMOGRAPHIC GROUPS FOR DECISION MAKERS

To alleviate concerns about the simplicity of the experimental setup and further validate our findings, we conducted a follow-up experiment replicating our original analysis on various demographic profiles with varying age and gender (Tables 14 and 15). We find that our analyses stay consistent across all these factors, suggesting that our findings are at least somewhat robust to variations in context and prompt, as well as differing profiles of people.

## E.3 LESS RATIONAL DECISION MAKERS

For additional prompt variation, we also prompt the LLM to make the same predictions for a monkey rather than a human (Tables 14 and 15, bottom row). When performing both zero-shot and CoT, models accurately reflect that monkeys are less rational and the resulting predictions are less correlated with max expected value. However, for the CoT condition, the correlations are also increased with actual human behavior. This supports our conclusions with an example where the model predicts less rational decisions, but these are actually more in-line with how humans actually behave.

## F FULL RESULTS FOR INVERSE MODELING

In this section, we provide the correlation plots between humans/rational models and LLMs. For brevity, we consider only the positive CoT context. The ordering of the plots begins with GPT-4o, followed by GPT-4 Turbo, Claude-3 Opus, Llama-3-70B, and concludes with Llama-3-8B.

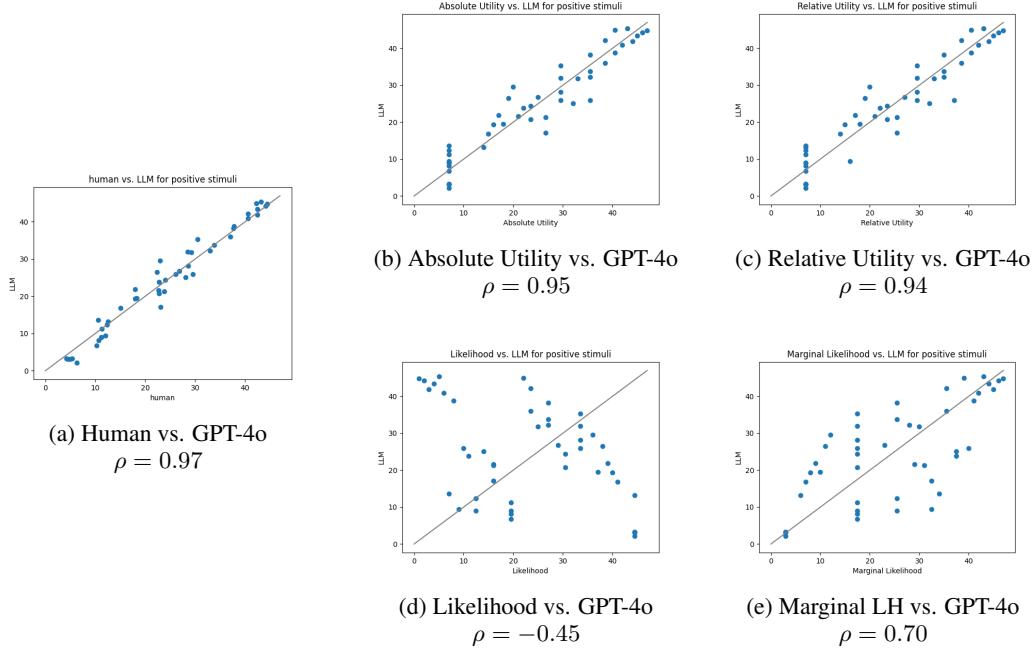


Figure 3: Comparing GPT-4o CoT rankings (y-coordinates) to humans and four theoretical decision-making models (x-coordinates) in positive setting.

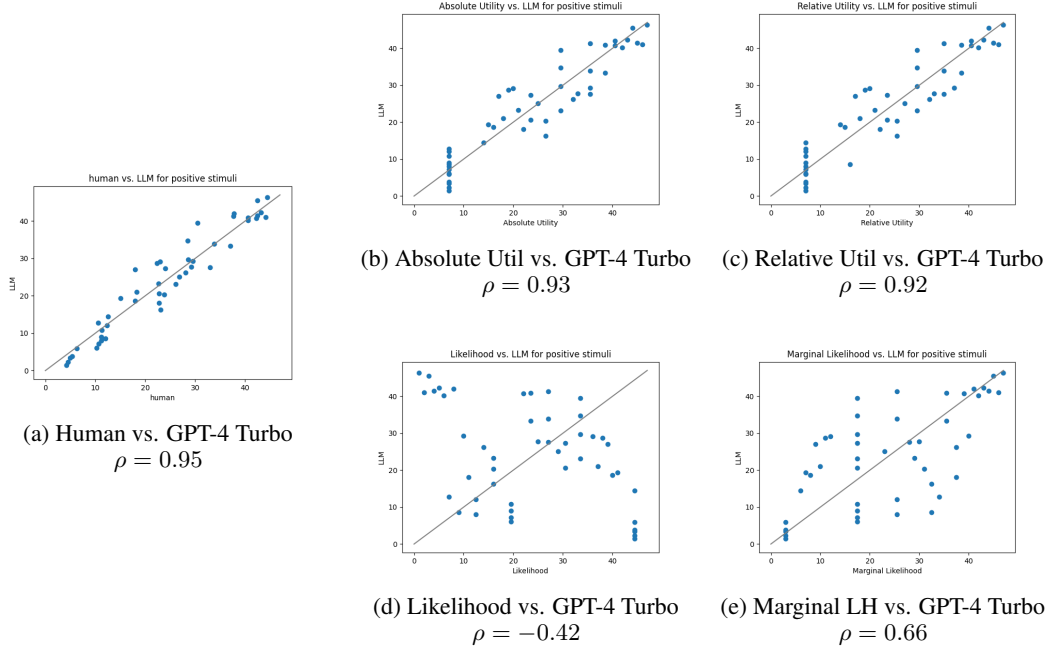


Figure 4: Comparing GPT-4 Turbo (0125-preview) CoT rankings (y-coordinates) to humans and four theoretical decision-making models (x-coordinates) in positive setting.